Ericsson PE Node Failure Resolution SOP for JAY-PE-09 at Jayanagar Metro Site

Node: JAY-PE-09 | **Vendor:** Ericsson | **Severity:** Warning | **Alarm Code:** ALARM_6443

Description

The failure domain of Hardware/Environmental Fault on the Ericsson PE node JAY-PE-09 at the Jayanagar Metro Site is attributed to the occurrence of ASIC memory parity errors or line card crashes, resulting in the loss of power supply redundancy. This is a critical failure scenario that necessitates immediate attention to prevent network downtime and ensure continuity of service. In the Ericsson PE node, the ASIC memory parity errors are detected by the system's built-in error correction mechanisms, which trigger an alarm (ALARM_6443) to alert the network administrators. The line card crashes, on the other hand, can be caused by a combination of factors, including power supply issues, environmental stress, or hardware failures. The loss of power supply redundancy further exacerbates the situation, making it essential to identify and rectify the root cause of the failure as quickly as possible.

Pre-requisite Checks

- Ensure remote hands are dispatched to the Jayanagar Metro Site for physical access and assistance.

- Verify the status of redundant links to ensure they are operational and not contributing to the failure scenario.

- Check the power supply redundancy status to confirm that it is functioning correctly.

Potential Root Causes

Cause	Probability
ASIC memory parity errors	High
Line card crashes	Medium
Power supply redundancy loss	Low

Diagnostic Commands & Expected Outputs

```
Command: admin:System/show system status, Expected Output: System Status: OK, Description: This command is used to verify the overall system status, including the health of the ASIC memory and the line cards.

Command: admin:System/show memory status, Expected Output: Memory Status: OK, Description: This command is used to check the status of the ASIC memory, including any detected errors or issues.

Command: admin:System/show line card status, Expected Output: Line Card Status: OK, Description: This command is used to verify the status of the line cards, including any detected errors or issues.
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```
Command: admin:System/show power supply status, Expected Output: Power Supply Status: OK, Description: This command is used to check the status of the power supplies, including any detected issues or redundancy loss.
```

Resolution Steps

Step 1: Step: Identify and replace faulty ASIC memory, Command: admin:System/replacement asic-memory, Description: This command is used to replace the faulty ASIC memory with a new one, which should resolve the memory parity errors.

Step 2: Step: Reboot the line card, Command: admin:System/reboot line-card, Description: This command is used to reboot the line card, which should resolve any crashes or issues detected.

Step 3: Step: Verify power supply redundancy, Command: admin:System/show power supply status, Description: This command is used to verify that the power supply redundancy is functioning correctly and not contributing to the failure scenario.

Post-Check Verification

```
Command: admin:System/show system status, Expected Output: System Status: OK, Description: This command is used to verify that the system is stable and functioning correctly after the resolution steps.

Command: admin:System/show memory status, Expected Output: Memory Status: OK, Description: This command is used to verify that the ASIC memory is functioning correctly and there are no detected errors or issues.

Command: admin:System/show line card status, Expected Output: Line Card Status: OK, Description: This command is used to verify that the line cards are functioning correctly and there are no detected errors or issues.

Command: admin:System/show power supply status, Expected Output: Power Supply Status: OK, Description: This command is used to verify that the power supplies are functioning correctly and there are no detected issues or redundancy loss.
```

Roll-back Procedures

- Step: Revert changes made during resolution, Command: admin:System/revert changes, Description: This command is used to revert any changes made during the resolution steps, ensuring that the system is restored to its original state.

- Step: Reinstall the original components, Command: admin:System/install original-components, Description: This command is used to reinstall the original components, ensuring that the system is restored to its original state.